

# Why Robots Are Untaxed?

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## Abstract

This paper proposes a simple, static, partial equilibrium model of automation in which two representative firms from two countries engage in Bertrand-style price competition. The purpose of the model is to rationalise why many countries remain reluctant to implement tax on robots albeit the academia advices otherwise. The model illustrates that increase in tax on robots raises marginal cost, weakens price competitiveness, and reduces the demand for the products. To overcome such a loss in competitiveness, strategic interaction across countries leads to a Nash equilibrium with zero tax on robots.

**Keywords:** Automation, Robots, Bertrand Competition, Tax Competition.

**JEL Classifications:** H21, O33.

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# 1 Introduction

Automation raises productivity, but also disrupts the labor market and displaces some jobs (Acemoglu and Restrepo, 2021). Several studies argue that the slowing down automation through taxing robots is optimal (Guerreiro et al., 2022; Costinot and Werning, 2022; Beraja and Zorzi, 2025). However, such proposals have gained limited policy traction in practice: No major economy has implemented a comprehensive direct robot tax. In the European Union (UE), for instance, robot tax initiative was rejected because they argue that it could have a negative impact on competitiveness and innovation.<sup>1</sup> Despite a widening deviation between academic and public debate, however, relatively little attention has been devoted to explaining why governments may be reluctant to adopt robot taxes.

This paper studies strategic tax competition on robots. I develop a partial equilibrium model of automation with two countries, two firms, and a homogeneous product in which firms engage in Bertrand-style price competition. This model is motivated by the firm-side component of the static general equilibrium model developed by Guerreiro et al. (2022). However, this paper departs from their setup by endogenising prices through Bertrand-style competition. Derived from the optimal price competition, the model then characterises the Nash equilibrium of strategic tax competition between two countries.

My model suggests that a higher robot tax increases marginal cost. When one country imposes a higher tax on robots than its competitor, domestic firm loses price competitiveness in international market. Such a loss in competitiveness reduces output and inputs demand. In contrast, the competing country gains positive externalities through improved relative competitiveness. As the government aims to overcome such a loss in competitiveness, strategic interaction across countries leads to a Nash equilibrium with zero tax on robots.

This paper contributes the current literature by highlighting price competition in international market. Although the model only incorporates with partial equilibrium, this paper establishes competition mechanism which is absent in Guerreiro et al. (2022).

This paper is organised as the following sections. Section 2 elaborates on the economic model. Section 3 concludes the paper.

## 2 Model

### 2.1 Environment

The model incorporates with two countries  $i = \{A, B\}$ , which are assumed to be identical. In particular, each country has a representative firm with identical technology and input prices. The representative firms from country  $A$  and  $B$  produce a homogeneous product that is sold in the global market. The total world demand is normalised to unity such that  $1 = Y_A + Y_B$  where  $Y_A \in (0, 1)$ .

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<sup>1</sup><https://www.reuters.com/article/world/uk/european-parliament-calls-for-robot-law-rejects-robot-tax-idUSKBN15V2K9/>

## 2.2 Hotelling Demand

The total world demand is generated by a continuum of importing-countries  $m$  with unit mass which are equally distributed along a geographic line spectrum  $z_m \in [0, 1]$ . Let  $p_A$  and  $p_B$  are the prices set by the representative firms from country  $A$  and  $B$ , respectively, and  $t$  is the unit cost of transportation. Each importing-country buys one unit of product either from country  $A$  or  $B$  based on the following decision rule:

- (i) Importing-country  $m$  buys from country  $A$  if  $p_A + t(z_A - z_m)^2 < p_B + t(z_B - z_m)^2$
- (ii) Importing-country  $m$  buys from country  $B$  if  $p_A + t(z_A - z_m)^2 > p_B + t(z_B - z_m)^2$
- (iii) Importing-country  $m$  is indifferent between buying from country  $A$  or  $B$  if  $p_A + t(z_A - z_m)^2 = p_B + t(z_B - z_m)^2$

According to the decision rule, we can determine the total demand for product from country  $A$  by the value of  $z$  such that importing-country  $\tilde{m}$  is indifferent (i.e.,  $z_{\tilde{m}} \equiv Y_A$ ). Let country  $A$  be located at  $z_A = 0$  while the country  $B$  at  $z_B = 1$ , then the total (Hotelling) demand for country  $A$  is defined as follows.

$$Y_A = \frac{1}{2} - \left( \frac{p_A - p_B}{2t} \right) \quad (1)$$

Equation 1 suggests that if the representative firms from both countries set the same prices, the demand will be divided in half. Otherwise, the demand for product produced by country  $A$  will be larger than half if  $p_A$  is lower than  $p_B$ , *vice versa*.

## 2.3 Firm

In country  $i$ , robots are produced by competitive firms and it costs  $\phi$  a unit. A representative robot-producing firm maximises profit  $p_x X_i - \phi X_i$  by choosing the supply of robots  $X_i$ . The equilibrium implies  $p_x = \phi$ . The representative final good producer hires automated labor  $N_{r,i}$  and non-automated workers  $N_{n,i}$  and buys robots  $X_i$  supplied by the representative robot-producing firm. Following Guerreiro et al. (2022), the production function is given by

$$Y_i = A_i \underbrace{\left( \int_0^v \omega_i x_{ij}^{\frac{\rho-1}{\rho}} dj + \int_v^1 \iota_i n_{ij}^{\frac{\rho-1}{\rho}} dj \right)^{\frac{\rho}{1-\rho}(1-\alpha)}}_{\text{automated tasks}} \underbrace{N_{n,i}^\alpha}_{\text{non-automated tasks}} \quad (2)$$

where  $\alpha \in (0, 1)$ ,  $\rho \in [0, \infty)$ .  $v \in [0, 1]$  is the degree of automation.  $A_i$  is Total Factor Productivity (TFP).

As illustrated by Guerreiro et al. (2022), by assuming interior solution  $v \in (0, 1)$  and  $\omega_i = \iota_i$  for all  $i$  (i.e., robots and routine workers are equally productive), Equation 2 can be simplified to,

$$Y_i = A_i (X_i + N_{r,i})^{(1-\alpha)} N_{n,i}^\alpha \quad (3)$$

The representative firm maximises profit by choosing non-automated workers  $N_n$ , automated workers  $N_r$ , robots  $X$ , and price  $p_i$  subject to technological constraint (3) and world demand (1). The Lagrange-form of the firm's profit maximisation problem is given as follows.

$$\max_{p_i, N_{r,i}, N_{n,i}, X_i} p_i Y_i - W_{r,i} N_{r,i} - W_{n,i} N_{n,i} - (1 + \tau_i^x) \phi X_i + mc_i (A_i (X_i + N_{r,i})^{(1-\alpha)} N_{n,i}^\alpha - Y_i)$$

where  $\tau_i^x$  denotes linear tax on robots and  $mc_i$  is Lagrange multiplier which is equivalent to marginal cost. By taking first-order condition with respect to inputs  $N_n$ ,  $N_r$ , and  $X$ , the marginal cost is defined as follows.

$$mc_i = \frac{(1 + \tau_i^x) \phi}{(1 - \alpha) A_i} \left( \frac{(1 - \alpha) W_{n,i}}{\alpha W_{r,i}} \right)^\alpha \quad (4)$$

From equation 4, we can infer that increases in tax on robots  $\tau^x$  proportionally increase the marginal cost  $mc_i$ . Note that this result serves as an important foundation for the main results of the paper.

## 2.4 Bertrand Price Competition

The first-order condition of firm's profit maximisation problem with respect to price  $p_i$  gives us the best response strategy  $p_i^*$  under Bertrand price competition as the following equation (see Appendix A.1 for the details).

$$p_i^* = \frac{t + p_{-i} + mc_i}{2} \quad (5)$$

Given the best response, the Nash equilibrium of Bertrand price competition is

$$p_i^* = t + \left( \frac{2mc_i + mc_{-i}}{3} \right) \quad (6)$$

Equation 6 implies that when  $mc_i = mc_{-i}$ , then the Nash equilibrium of prices is simply  $p_i = t + mc_i$ .

## 2.5 Tax on Robots and Output

In this section, I consider two cases where  $\tau_A^x = \tau_B^x = 0$  and  $\tau_A^x > 0, \tau_B^x = 0$ . The differences in output of country  $A$  drawn from the two cases can be inferred as impact of tax on robots on output.

**Case 1.** When the governments from both countries set  $\tau_A^x = \tau_B^x = 0$ , then the marginal costs level are equal,  $mc_A = mc_B$ . The price equilibrium is thus given by  $p_A^* = p_B^* = t + mc$  (i.e.,  $NE_1$  in Figure 1). According to equation 1, then output demands for both countries are split in half.

$$Y_A = Y_B = \frac{1}{2} \quad (7)$$

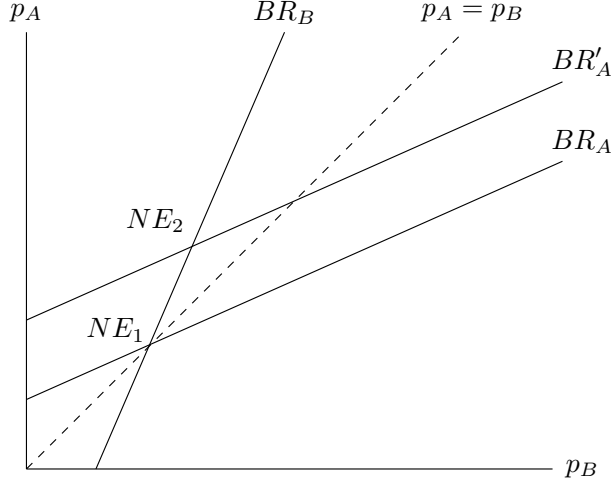


Figure 1: Nash Equilibrium of Price Competition

**Case 2.** Now let's consider a case where government of country  $A$  sets  $\tau_A^x > 0$  while the tax on robots of country  $B$  stays at  $\tau_B^x = 0$ . Non-zero robot tax increases marginal cost (as in equation 4), then the new marginal cost of country  $A$  is  $mc'_A > mc_B$ . Given this situation, the Nash equilibrium of price competition becomes

$$(p_A^*, p_B^*) = \left( t + \frac{mc_B + 2mc'_A}{3}, t + \frac{2mc_B + mc'_A}{3} \right) \quad (8)$$

Since  $mc'_A > mc_B$ , then  $p_A^* > p_B^*$ . The output of country  $A$  is therefore defined as follows.

$$Y'_A = \frac{1}{2} - \left( \frac{mc'_A - mc_B}{6t} \right) \quad (9)$$

To examine the impact on output, let  $\kappa_A$  be the ratio of  $Y'_A$  over  $Y_A$ .

$$\kappa_A = 1 - \frac{mc'_A - mc_B}{3t} < 1 \quad (10)$$

In this regard,  $\kappa_A$  can be seen as TFP-equivalent shocks of non-zero tax on robots on output  $Y'_A = \kappa Y_A$ . Given that the production function is homogeneous, then such a shock depresses inputs equally.

$$Y'_A = A(\underbrace{\kappa_A X_A}_{X'_A < X_A} + \underbrace{\kappa_A N_{r,A}}_{N'_{r,A} < N_{r,A}})^{(1-\alpha)} (\underbrace{\kappa_A N_{n,A}}_{N'_{n,A} < N_{n,A}})^\alpha \quad (11)$$

On the other hand, country  $B$  gains positive externalities from higher marginal cost in country  $A$  due to non-zero tax on robots. The TFP-equivalent shocks of non-zero tax on robots for country  $B$  is  $\kappa_B > 1$ . Therefore,

$$Y'_B = A(\underbrace{\kappa_B X_B}_{X'_B > X_B} + \underbrace{\kappa_B N_{r,B}}_{N'_{r,B} > N_{r,B}})^{(1-\alpha)} (\underbrace{\kappa_B N_{n,B}}_{N'_{n,B} > N_{n,B}})^\alpha \quad (12)$$

**Proposition 1.** Suppose that  $i \in \{A, B\}$ . If  $\tau_i^x > \tau_{-i}^x$  then country  $i$  suffers from output contraction by  $\kappa_i - 1 < 0$ . Given that the production functions from both countries are homogeneous, then the number of  $X_i$ ,  $N_{n,i}$ , and  $N_{r,i}$  also decline by  $\kappa_i - 1$ , while country  $-i$  gains positive externalities by  $\kappa_{-i} - 1 > 0$ .

## 2.6 Nash Equilibrium of Non-cooperative Tax Competition

Suppose governments choose  $\tau_i^x \in \mathbb{R}_+$  to maximise domestic output due to, for instance, political incentives associated with employment and industrial competitiveness. Based on **Proposition 1**, if country  $i$  sets tax on robots  $\tau_i^x$  lower than  $\tau_{-i}^x$ , then they receive positive shock  $\kappa_i - 1 > 0$ . In contrast, country  $B$  experiences negative externalities  $\kappa_{-i} - 1 < 0$ . In response country  $B$  will lower its tax  $\tau_{-i}^x$ . This leads to race-to-the-bottom tax competition between country  $i$  and  $-i$ . As such, the Nash Equilibrium of tax competition is  $\tau^x = 0$ .

**Corollary 1.** If the governments are output maximising by choosing  $\tau_i^x \in \mathbb{R}_+$ , then Nash equilibrium of tax competition is  $\tau_i^x = 0$  for all  $i \in \{A, B\}$ .

*Proof.* See Appendix A.2 for formal proof. □

## 3 Conclusion

Why many countries remain reluctant to implement tax on robots albeit the academia advises otherwise? This paper argues that a higher robot tax could lead to loss in relative competitiveness, causing lower output and demand for labour inputs. Under special case where the government maximises domestic output, the Nash equilibrium of tax competition is characterised by zero tax on robots.

However, the zero tax equilibrium characterised in this paper does not necessarily imply social optimal outcomes. While zero tax equilibrium preserves competitiveness, non-zero tax also can cushion the adverse effects of automation on labour market as illustrated by Guerreiro et al. (2022).

From this perspective, the general equilibrium may resemble a prisoner's dilemma. Unilateral taxation weakens competitiveness and causes race-to-the-bottom tax competition. However, coordinated taxation through international cooperation may eliminate loss in relative competitiveness and improve welfare outcomes for all countries labour market protection. Therefore, extending the framework to general equilibrium model with welfare-maximising governments can be a potential future research.

## References

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## A Appendices

### A.1 Best Response Strategy and Price Equilibrium

Firm sets price to maximise their profit subject according this problem:

$$\max_{p_i} p_i Y_i - mc_i Y_i$$

Subject to Hotelling demand in equation 1,

$$Y_i = \frac{1}{2} - \frac{p_i - p_{-i}}{2t}$$

The first-order condition with respect to  $p_i$ , which is the best response strategy, is therefore given by

$$p_i^* = \frac{t + p_{-i} + mc_i}{2} \quad (\text{A.13})$$

Nash equilibrium of price competition is defined as follows.

$$p_i^* = \frac{t + p_{-i}^* + mc_i}{2} \iff p_i^* = \frac{t + \left( \frac{t + p_i^* + mc_{-i}}{2} \right) + mc_i}{2}$$

Therefore,

$$p_i^* = t + \left( \frac{mc_{-i} + 2mc_i}{3} \right) \quad (\text{A.14})$$

### A.2 Proof of Nash Equilibrium of Tax Competition

Let countries be otherwise identical and suppose that the governments maximise  $Y_i(\tau_i^x, \tau_{-i}^x)$  by choosing  $\tau_i^x \in \mathbb{R}_+$ . From Proposition 1, if country  $i$  sets a higher robot tax than its competitor,  $\tau_i^x > \tau_{-i}^x$ , then its marginal cost becomes higher than that of country  $-i$

$$mc_i > mc_{-i}$$

This weakens country  $i$ 's price competitiveness and reduces its output:

$$Y_i(\tau_i^x, \tau_{-i}^x) < Y_i(\tau_{-i}^x, \tau_{-i}^x)$$

Conversely, if country  $i$  sets a lower robot tax than its competitor,  $\tau_i^x < \tau_{-i}^x$ , then country  $i$  obtains a lower marginal cost than country  $-i$ :

$$mc_i < mc_{-i},$$

which improves its relative price competitiveness and increases its output:

$$Y_i(\tau_i^x, \tau_{-i}^x) > Y_i(\tau_{-i}^x, \tau_{-i}^x)$$



Therefore, for any given tax rate chosen by country  $-i$ , country  $i$  maximises output by choosing the lowest feasible robot tax. Since taxes are restricted to be non-negative,  $\tau_i^x \in \mathbb{R}_+$ , the best response of country  $i$  is

$$BR_i(\tau_{-i}^x) = 0$$

The same argument applies symmetrically to country  $-i$ , so

$$BR_A(\tau_B^x) = 0, \quad BR_B(\tau_A^x) = 0$$

The Nash equilibrium is the pair of tax rates where both countries are playing mutual best responses. Hence,

$$(\tau_A^{x*}, \tau_B^{x*}) = (0, 0)$$

Since zero is the unique best response for each country for any  $\tau_{-i}^x \in \mathbb{R}_+$ , the Nash equilibrium is unique.